# A lightweight aluminum profile surface defect detection algorithm based on the improved YOLOv8 algorithm

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## Abstract

Due to imperfect manufacturing processes and external factors, aluminum profiles can exhibit various surface defects, which significantly affect their service life. To address the issues of low accuracy and high computational load in current aluminum profile surface defect detection methods, this paper proposes a high-precision and lightweight aluminum profile surface defect detection model, YOLOv8-ARL. The model first replaces some Conv modules in the backbone and neck with an Adown down-sampling module, which reduces the number of parameters while retaining more image information. Next, a lightweight module, RGCSPELAN (RepGhostC-SPELAN), is introduced to replace the C2f module in the network, enhancing gradient flow information while improving computational resource utilization. Finally, a lightweight detection head is constructed using the concept of shared convolution, replacing the BN (Batch Normalization) layers in the Conv modules to ensure that lightweight design does not compromise accuracy. The improved YOLOv8-ARL algorithm achieves an average mean precision (mAP50%) of 88.6% on the APDDD aluminum profile dataset, a 2.2% increase over the original YOLOv8s model, with a parameter count of 4.55M, reduced by 59% compared to the original model, and a computational load reduced by 48%. The detection time per image is 2.2ms. Experimental results demonstrate that the improved YOLOv8-ARL model maintains excellent detection accuracy and speed while being lightweight.

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## CCS Concepts

• **Computing methodologies** → Machine learning; Machine learning algorithms; Feature selection.

## Keywords

Defect detection, YOLOv8 algorithm, Adown module, RGCSPELAN module, lightweight detection head

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## 1 Introduction

In practical production settings, aluminum profiles can develop various surface defects due to factors such as uneven alloy composition, improper temperature control during extrusion, mold wear, and inadequate subsequent surface treatment. These issues not only affect the aesthetic appeal of aluminum profiles but also compromise the material's durability and load-bearing capacity, ultimately diminishing product quality. Therefore, efficient and accurate detection of surface defects during production is crucial for ensuring the safety and longevity of aluminum profiles across different applications.

Currently, deep learning-based object detection methods are primarily categorized into two types: two-stage detection methods and one-stage detection methods. Two-stage detection methods, represented by R-CNN [1], Mask R-CNN [2], and Faster R-CNN [3], first generate candidate regions and then classify these regions and regress bounding boxes, usually achieving higher detection accuracy. Han Lu [4] and others enhanced the Mask R-CNN-based network structure by improving feature extraction and optimizing loss calculation methods, thereby increasing detection accuracy and efficiency on the pantograph carbon slider dataset. Liu Yingtao [5] proposed a defect detection method for wheel tread surfaces based on an improved Faster R-CNN. By incorporating a CBAM attention mechanism and a K-means++ clustering-based anchor box scheme, the method achieved an average precision of 97.3%. However, the improved network's complexity is high due to the presence of numerous fully connected layers. Despite the commendable accuracy of two-stage networks, they often come with large parameter counts and computational loads, and their detection speed may not meet the demands of real-world industrial production.

One-stage detection methods predict both category and bounding box positions directly within the network without separately generating candidate regions, resulting in faster detection speeds. Examples include SSD [6] and the YOLO [7] series. Li Wei [8] proposed an improved YOLOv7-based steel surface defect detection algorithm, incorporating GhostConv lightweight modules and global attention mechanisms to effectively enhance detection accuracy. However, the parameter count remains large at 43.6MB. Su Jia [9] designed an efficient large convolution module and multi-branch parallel feature fusion module based on the original YOLOv8 model. This model achieved good detection results on the GC10-DET and DeepPCB datasets, with a parameter count of only 2.7MB. While one-stage algorithms have become mature in industrial defect detection, their accuracy, parameter count, and computational load still need improvement for specific types of defects and complex detection environments, such as aluminum profile defect detection. To address these issues, this paper proposes a high-precision, lightweight aluminum profile surface defect detection algorithm based on an improved YOLOv8 model, referred to as YOLOv8-ARL.

## 2 YOLOv8-ARL: A Lightweight Surface Defect Detection Model for Aluminum Profiles

YOLOv8, introduced in January 2023 by Ultralytics, the developers of YOLOv5, is a single-stage detection model. The backbone of YOLOv8 incorporates a new C2f module, replacing the previous C3 module, which reduces computational load while enhancing gradient flow. The neck of YOLOv8 eliminates the 1×1 down-sampling layer present in YOLOv5 but retains the PAN [10] concept. In the head structure, YOLOv8 shifts from an Anchor-Based to an Anchor-Free approach and adopts the current mainstream Decoupled-Head structure [11], which separates classification and detection heads. It introduces VFL Loss as the classification loss, combined with DFL Loss and CIoU Loss for regression. These improvements result in enhanced accuracy and speed compared to earlier versions.

Considering both detection accuracy and model size, this paper proposes YOLOv8-ARL, a lightweight aluminum profile surface defect detection algorithm, based on YOLOv8s. The network architecture of YOLOv8-ARL is illustrated in Figure 1. The main improvements are as follows:

1). Adown Down-Sampling Module: Some Conv modules in YOLOv8 are replaced with the Adown down-sampling module proposed in YOLOv9 [12]. This module employs max pooling and Conv layers for down-sampling, optimizing parameter quantity while preserving the integrity and diversity of feature map information.

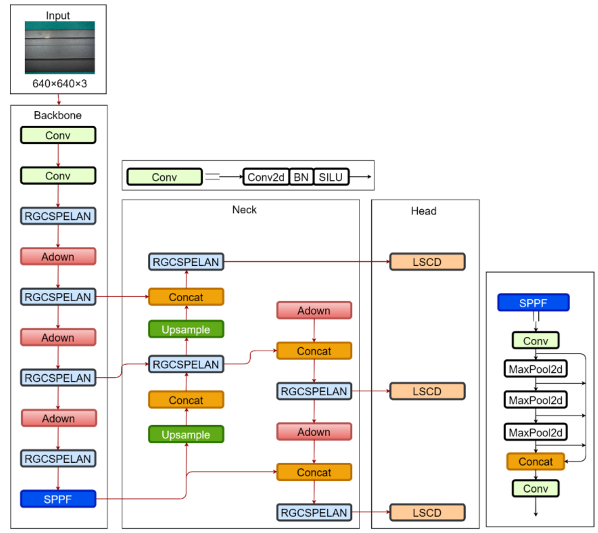


**Figure 1: YOLOv8-ARL algorithm structure diagram.**

The Adown module can be flexibly placed at various locations in the model. In this work, it replaces the 3rd, 4th, and 5th Conv modules in the original YOLOv8 backbone, as well as all Conv modules in the neck.

2). RGCSPELAN Module: A new lightweight RGCSPELAN (RepGhostCSPELAN) module, designed by integrating the concepts of CSPNet [13] and ELAN, is introduced to replace the C2f module. This module ensures a richer gradient flow while maintaining lightweight characteristics and allows for adjustable feature extraction capabilities.

3). Lightweight Detection Head: A lightweight detection head is constructed using the concept of shared convolution. The BN (Batch Normalization) layers are replaced to compensate for the accuracy loss associated with the lightweight design.

### 2.1 RGCSPELAN Module

In YOLOv8, the C2f module is utilized for feature transformation and fusion. This module is a CSP (Cross Stage Partial) Bottleneck with two convolutional layers, primarily designed to introduce cross-stage connections within residual blocks. Observing the parameter count during the training of YOLOv8s, it is noted that the C2f module accounts for more than half of the entire model's parameters, which increases the model's hardware requirements and computational cost. To address this issue and better meet the needs of mobile platforms, this paper introduces the RGCSPELAN module, designed by integrating the concepts of CSPNet and ELAN to replace the C2f module. The structure of the RGCSPELAN module is illustrated in Figure 2.

First, a convolutional layer is applied to transform the input feature map and adjust its channel dimensions. The feature map is then split along the channel dimension into two branches. The first branch is passed directly to the output, while the second branch undergoes a series of computational operations.

The RGCSPELAN module discards the commonly used BottleNeck layer in YOLOv8, which inevitably leads to a reduction in

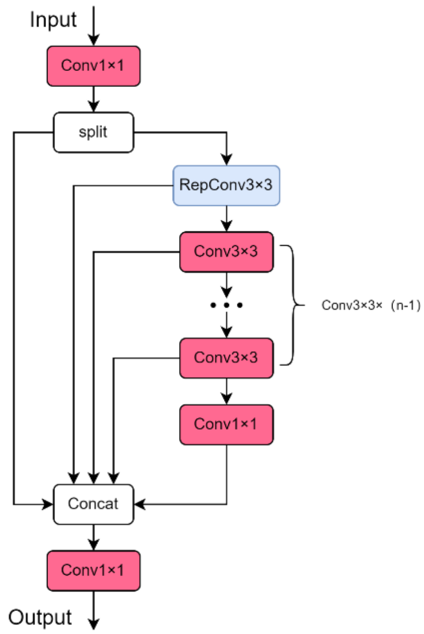


**Figure 2: RGCSPELAN module structure diagram.**

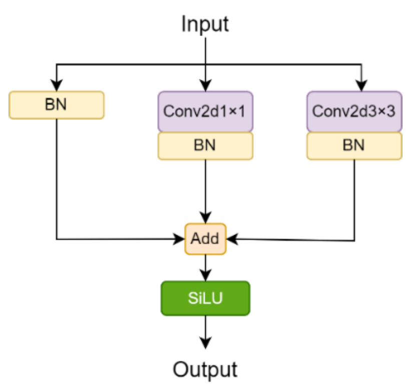


**Figure 3: RepConv structure diagram.**

precision despite the decrease in parameter count. To mitigate this, RepConv [14] is introduced in the second branch for feature extraction. The RepConv design, as illustrated in Figure 3, employs a multi-branch convolutional layer structure that enables the model to learn diverse features simultaneously, thus enhancing feature extraction capability.

In the second branch, RepConv is used initially to extract features, improving both feature extraction ability and gradient flow. The output channels of RepConv are set to the input channels multiplied by a scaling factor q, allowing control over the feature extraction capacity of the RGCSPELAN module by adjusting q. Following this, n-1 Conv modules are used for further feature extraction. Inspired by GhostNet [15], a 1×1 Conv module replaces multiple traditional convolution operations to generate redundant feature maps, providing a more comprehensive understanding of the input data while saving computational resources due to the cost-effective nature of this operation.

Subsequently, all feature maps from the two branches are concatenated, merging information from different branches and thereby enriching feature representation. Finally, a 1×1 Conv module adjusts the channel dimensions. The proposed RGCSPELAN module, compared to the C2f module, achieves a reduction in both parameter count and computational complexity while providing richer gradient flow information. This enhancement effectively improves model detection accuracy and efficiency.

## 2.2 LSCD Detection Head

The YOLOv8 detection head employs a decoupled head structure, separating classification and localization tasks. Compared to conventional detection heads, the decoupled head offers superior feature representation and learning capabilities, as well as accelerated model convergence. However, as indicated by the parameter count for YOLOv8, the detection head of YOLOv8s accounts for nearly one-fifth of the entire model's parameters. To address this issue, this paper proposes a lightweight detection head, LSCD (Lightweight Shared Convolutional Detection Head), which utilizes shared convolutions.

The LSCD module processes feature maps of resolutions 80×80, 40×40, and 20×20 to accommodate defects of varying sizes. Initially, three 1×1 convolutional modules are used to adjust the channels of the input feature maps. Subsequently, inspired by the concept of shared convolutions, the three detection heads share two 3×3 convolutional modules for feature extraction. For classification tasks, irrespective of the size of the detection layer (large, medium, or small), the classification target remains consistent. Therefore, a shared Conv2d convolution is employed for classification. To address the issue of varying target scales in localization tasks, each detection head uses a shared convolution Conv2d×scale for feature scaling, where scale is a scaling factor with an initial value of 1. Through backpropagation, each detection head learns an optimal scale value.

However, the use of shared convolutions for lightweight design may lead to diminished feature extraction capability. The Conv module consists of standard 2D convolutional layers, BN (Batch Normalization) layers, and SiLU activation functions. It has been observed that BN layers may not accurately estimate the mean and variance of the current batch when the batch size is small, especially in low-computation devices where a large batch size cannot be set. Group Normalization (GN), which normalizes features within groups of channels, effectively mitigates this issue. Previous research [16] has demonstrated that GN layers in the head structure can improve classification and localization performance. Therefore, BN layers in all Conv modules of the proposed head structure are replaced with GN layers, referred to as Conv_GN modules, to compensate for the accuracy loss due to the lightweight design. The structure of the LSCD module is illustrated in Figure 4.

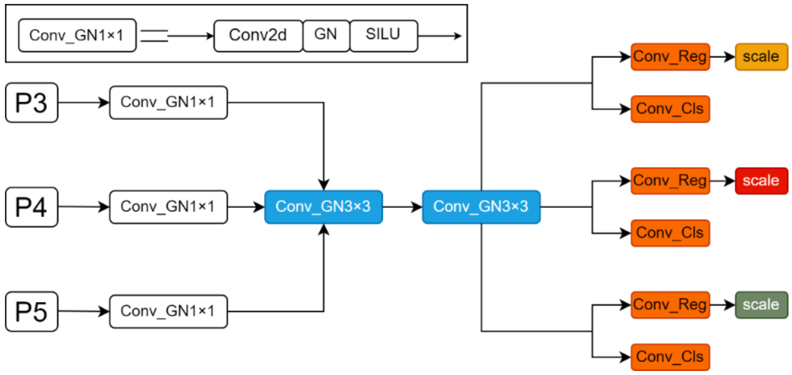


**Figure 4: LSCD detection head structure diagram.**

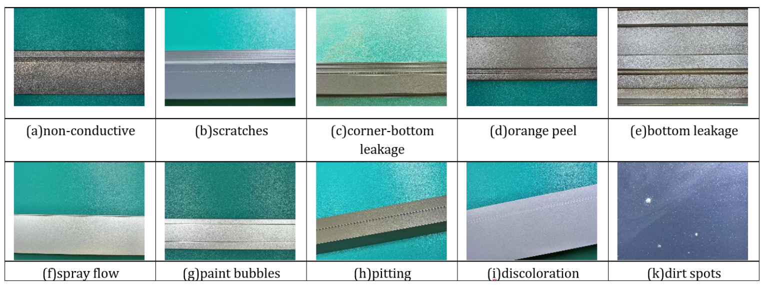


**Figure 5: Aluminum profile surface defect sample.**

**Table 1: Aluminum profile data set defect data statistics.**

| types of defects | non-conductive | scratches | corner-bottom leakage | orange peel | bottom leakage | spray flow | paint bubbles | pitting | discoloration | dirt spots |
|---|---|---|---|---|---|---|---|---|---|---|
| Before expansion | 388 | 128 | 69 | 173 | 538 | 86 | 82 | 407 | 365 | 261 |
| After expansion | 388 | 278 | 345 | 346 | 538 | 410 | 308 | 407 | 365 | 261 |

# 3 Experimental Design and Results Analysis

## 3.1 Experimental Dataset

This study utilizes the Tianchi Aluminum Profile Defect Detection Dataset (APDDD) to validate the effectiveness of the proposed algorithm. The dataset includes ten types of defects: non-conductive, scratches, corner-bottom leakage, orange peel, bottom leakage, spray flow, paint bubbles, pitting, discoloration, and dirt spots. It comprises a total of 2,497 images, all collected in real production environments. The images are stored in JPG format, with dimensions of 2560×960 pixels. Representative defect samples are shown in Figure 5.

After analyzing the sample sizes in the dataset, it was observed that the number of samples for defects such as scratches, corner-bottom leakage, orange peel, spray flow, and paint bubbles was significantly smaller compared to other defects. To prevent sub-optimal model performance due to imbalanced class distribution, this study augmented the dataset using the albumentations library. The augmentation techniques included vertical flipping, horizontal flipping, brightness adjustment, rotation, and random scaling. The dataset statistics before and after augmentation are shown in Table 1, demonstrating a more balanced dataset after augmentation. Given that high image resolution consumes considerable computational resources and training time, the images were resized to 1280×960 pixels. The MVTec Deep Learning Tool software was used to annotate the dataset. The dataset was finally split into training, validation, and test sets with a ratio of 7:1:2, resulting in 2,536 images for training, 364 for validation, and 726 for testing.

## 3.2 Experimental Environment Configuration and Evaluation Metrics

The training parameters are detailed in Table 2. All experiments were conducted under identical conditions to ensure the reliability of the results. The experimental environment parameters are as follows: CPU - Intel i9-13900K, RAM - 128GB, GPU - NVIDIA RTX4090 x2 with 24GB of VRAM, operating system - Ubuntu 20.04, and deep learning framework - PyTorch 2.0.1 with CUDA 11.8.

**Table 2: Training parameters.**

| Parameter Name | Parameter Value |
|---|---|
| Learning Rate | 0.01 |
| Data Augmentation Techniques | close_mosaic=10 |
| Total Training Epochs | 500 |
| Batch Size | 16 |
| Optimizer | SGD |
| Optimizer Momentum | 0.937 |
| Weight Decay Coefficient | 0.0005 |

To evaluate the overall performance of the model, this study employs the following metrics: Average Precision (AP), Mean Average Precision at IoU threshold 0.5 (mAP@0.5), inference time per image (T), number of parameters (Params), and computational complexity (FLOPs).

## 3.3 Experimental Analysis of the RGCSPELAN Module

The RGCSPELAN module is implemented by replacing all C2f modules in YOLOv8 with the RGCSPELAN module. This module incorporates a scaling factor q, which adjusts the output channel number of RepConv and subsequently changes the input channel number of the following Conv module. Although the input channel number does not directly affect the spatial dimensions, adjusting it can indirectly modify the granularity of information processing. Using more input channels allows for capturing richer feature information but also increases the algorithm's parameter count. To investigate the impact of q on the model's detection performance, parameter count, and computational load, and to select a relatively suitable value for q, the value of q is constrained within the range [0,1]. For efficiency in experimentation,q is discretized and tested at values of 0.25, 0.5, 0.75, and 1, with experiments conducted on a designated test set. The results of the experiments are presented in Table 3.

The YOLOv8s model with the introduction of the RGCSPELAN module shows notable improvements over the original model. Among the tested configurations, when q = 0.75, the Mean Average Precision at IoU threshold 0.5 (mAP@0.5) achieves the highest value of 87.5%. This indicates that the average precision does not necessarily increase with the number of channels. For instance, when q = 1.00, the mAP@0.5 decreases to 87.1%, likely due to the excessive channels extracting a greater amount of irrelevant feature information. When q = 0.25, the model has the fewest channels, resulting in the smallest parameter count of 7.20M, which is 35.1% less than the original model, and the lowest computational load of 18.6G, reduced by 34.7% compared to the original model. However, the mAP@0.5 for q=0.25 is only 86.8%. For both q=0.25 and q=0.5, the detection time per image is minimized to 2.1 ms, which is 0.4 ms faster than the original model. Considering these factors, using the RGCSPELAN module with q = 0.50 to replace the C2f module in the original model achieves a good balance between parameter count and accuracy. The resulting model attains an mAP@0.5 of 87.3%, with a parameter count of 7.90M and a computational load of 20.3G. Thus, the integration of a lightweight module not only reduces parameter count and computational load but also enhances detection speed.

## 3.4 Comprative Experiments

To further validate the effectiveness of the proposed YOLOv8-ARL algorithm, comparative experiments were conducted with other mainstream object detection network algorithms, including SSD, Faster R-CNN, DETR [17], RT-DETR [18], YOLOv5, YOLOv6 [19], YOLOv7 [20], YOLOv7-tiny, and YOLOv9. The experimental environment remained consistent with the previous sections, and the results are presented in Table 4.

Table 4 clearly shows that YOLOv8-ARL achieves optimal performance in terms of accuracy, parameter count, and detection speed, with computational load being only second to YOLOv7-tiny. The SSD algorithm's average mean precision is 38.2% lower than YOLOv8-ARL, and it also exhibits slower detection speeds with a computational load reaching 290G. Faster R-CNN achieves a precision of 71.5%, which represents a significant improvement over SSD but remains relatively low, and its model parameter count is as high as 41.39M. DETR and RT-DETR perform well in terms of precision, yet their parameter counts are 41.56M and 28.46M, respectively, and their computational loads are 81.6G and 100.6G, which imposes high resource requirements and complicates deployment. Among the YOLO series, YOLOv8s achieves the highest precision with an average mean precision of 86.4%, YOLOv7-tiny has the smallest model size with a parameter count of only 6.03M, and YOLOv10s has the fastest detection speed with a single image detection time of 2.3ms. YOLOv8-ARL outperforms these models in all three metrics: its average mean precision is 2.2% higher than YOLOv8s, its parameter count is 1.48M smaller than YOLOv7-tiny, and its single image detection time is 0.1ms shorter than YOLOv10s. In terms of computational load, YOLOv8-ARL is only marginally higher than YOLOv7-tiny at 14.9G compared to 13.1G. In summary, the proposed YOLOv8-ARL model demonstrates strong performance in both accuracy and lightweight design, laying a solid foundation for future research.

To provide a more intuitive comparison of the detection effects before and after the improvement, YOLOv8s and YOLOv8-ARL were evaluated on a pre-defined test set. Sample detection results are shown in Figure 6. The left and right columns display the detection results from YOLOv8s and YOLOv8-ARL, respectively. It is evident that the original model exhibited overlapping detection boxes for scratch defects, whereas the improved model accurately detected the location of defects without overlap. Additionally, the original model missed defects such as corner defects, spray marks, and dirt spots, failing to detect all instances. In conclusion, the improved YOLOv8-ARL effectively addresses issues of missed detections and overlapping detection boxes, providing better performance for detection tasks.

# 4 Conclusion

To address issues of low accuracy, large parameter count, and high computational complexity in aluminum profile surface defect detection models, this paper proposes a lightweight defect detection algorithm, YOLOv8-ARL. Built upon the YOLOv8s base model, YOLOv8-ARL introduces the Adown module to replace the original

**Table 3: Comparison test of different scaling factors q.**

| model | mAP@0.5 | Params | T | FLOPs |
|---|---|---|---|---|
| YOLOv8s | 86.4% | 11.12M | 2.5ms | 28.5G |
| YOLOv8s+RGCSPELAN(q=0.25) | 86.8% | 7.20M | 2.1ms | 18.6G |
| YOLOv8s+RGCSPELAN(q=0.50) | 87.3% | 7.90M | 2.1ms | 20.3G |
| YOLOv8s+RGCSPELAN(q=0.75) | 87.5% | 8.64M | 2.2ms | 22.1G |
| YOLOv8s+RGCSPELAN(q=1.00) | 87.1% | 9.43M | 2.2ms | 24.2G |

**Table 4: Experiments with other algorithm models.**

| model | mAP@0.5 | Params | T | FLOPs |
|---|---|---|---|---|
| SSD | 50.4% | 24.95M | 16.7ms | 290G |
| Faster-RCNN | 71.5% | 41.39M | 15.0ms | 178.0G |
| DETR | 83.5% | 41.56M | 9.5ms | 81.6G |
| RT-DETR | 86.4% | 28.46M | 5.0ms | 100.6G |
| YOLOv5s | 85.3% | 9.12M | 2.4ms | 23.8G |
| YOLOv6s | 84.9% | 16.3M | 2.8ms | 44.0G |
| YOLOv7 | 75.6% | 36.52M | 5.3ms | 103.3G |
| YOLOv7-tiny | 71.2% | 6.03M | 2.6ms | **13.1G** |
| YOLOv8s | 86.4% | 11.13M | 2.5ms | 28.5G |
| YOLOv9s | 83.9% | 9.60M | 14.5ms | 38.8G |
| YOLOv10s | 86.3% | 8.04M | 2.3ms | 24.5G |
| YOLOv8-ARL | **88.6%** | **4.55M** | **2.2ms** | 14.9G |

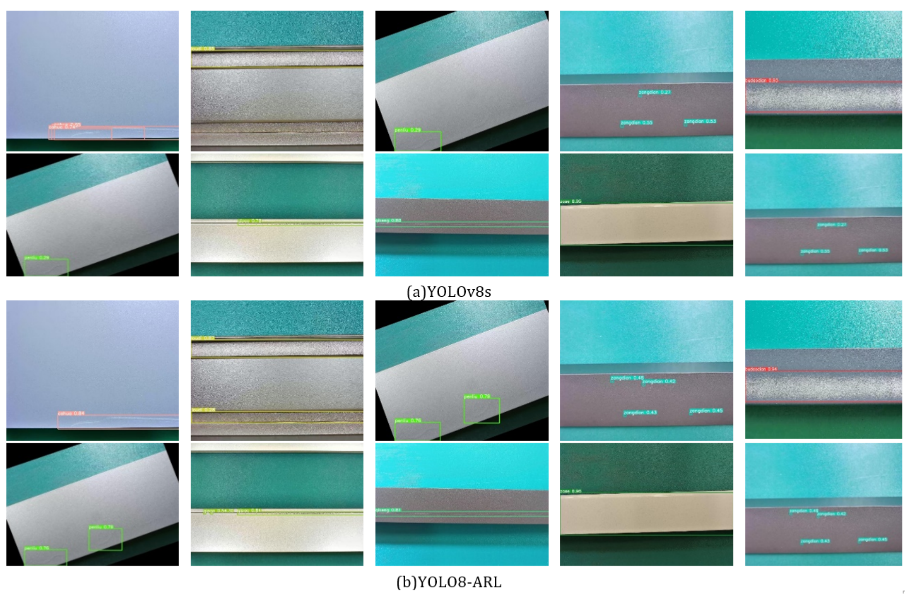


**Figure 6: Comparison of detection results.**

Conv module, thereby enhancing feature extraction capabilities while reducing the parameter count. Additionally, a lightweight RGCSPELAN module is designed to replace the C2f module in the original model, enriching gradient information flow and utilizing RepConv to compensate for any loss in accuracy while maintaining lightweight characteristics. The LSCD lightweight detection head is constructed, which uses shared convolutions to save substantial computational resources and incorporates a GN layer to improve detection accuracy.

Experimental results demonstrate that the proposed YOLOv8-ARL achieves notable performance on the APDDD aluminum profile dataset, with an average mean precision of 88.6%, a parameter count of only 4.55M, a computational load of 14.9G, and a single-image detection time of 2.2ms. These results indicate that the proposed model effectively addresses the challenges faced in aluminum profile surface defect detection. There is still room for further improvement in accuracy, and future work will focus on optimizing detection precision for specific defects such as scratches, paint bubbles, and dirt spots.

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## References

[1] Girshick, R., Donahue, J., Darrell, T. and Malik, J., 2014. Rich feature hierarchies for accurate object detection and semantic segmentation. In Proceedings of the IEEE conference on computer vision and pattern recognition (pp. 580-587).

[2] He, K., Gkioxari, G., Dollár, P. and Girshick, R., 2017. Mask r-cnn. In Proceedings of the IEEE international conference on computer vision (pp. 2961-2969).

[3] Ren, Shaoqing, Kaiming He, Ross Girshick, and Jian Sun. "Faster r-cnn: Towards real-time object detection with region proposal networks." Advances in neural information processing systems 28, 2015.

[4] Han Lu, Liu Taihao, Song Hailiang, *et al.* Optimal detection algorithm of pantograph carbon slide based on improved Mask R-CNN[J].Computer Applications and Software, 2024, 41(01):105-111+176.)

[5] LIU Yingtao, GUO Shiwei, FU Mengxin, *et al.* Wheelset Tread Defect Detection Based on Improved Faster RCNN[J]. Electronic Measurement Technology, 2024, 46(12): 34-41.

[6] Liu, W., Anguelov, D., Erhan, D., Szegedy, C., Reed, S., Fu, C.Y. and Berg, A.C., 2016. Ssd: Single shot multibox detector. In Computer Vision-ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11-14, 2016, Proceedings, Part I 14 (pp. 21-37). Springer International Publishing.

[7] Redmon, J. and Farhadi, A., 2017. YOLO9000: better, faster, stronger. In Proceedings of the IEEE conference on computer vision and pattern recognition (pp. 7263-7271).

[8] LI Wei, YAO Xiaomin, ZHANG Pengchao, *et al.* 2024. Research on improved YOLO-V7 steel surface defect detection algorithm[J/OL]. Mechanical Science and Technology: 1-10[2024-06-12].

[9] Su Jia, Jia Ze, Qin Yichang, *et al.* 2024. Improved YOLOv8 Algorithm for Industrial Surface Defect Detection[J/OL]. Computer Engineering and Applications, 1-15[2024-06-12].

[10] Liu, S., Qi, L., Qin, H., Shi, J. and Jia, J., 2018. Path aggregation network for instance segmentation. In Proceedings of the IEEE conference on computer vision and pattern recognition (pp. 8759-8768).

[11] Ge, Z., Liu, S., Wang, F., Li, Z. and Sun, J., 2021. Yolox: Exceeding yolo series in 2021. arXiv preprint arXiv:2107.08430.

[12] Wang, C.Y., Yeh, I.H. and Liao, H.Y.M., 2024. Yolov9: Learning what you want to learn using programmable gradient information. arXiv preprint arXiv:2402.13616.

[13] Wang, C.Y., Liao, H.Y.M., Wu, Y.H., Chen, P.Y., Hsieh, J.W. and Yeh, I.H., 2020. CSPNet: A new backbone that can enhance learning capability of CNN. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition workshops (pp. 390-391).

[14] Ding, X., Zhang, X., Ma, N., Han, J., Ding, G. and Sun, J., 2021. Repvgg: Making vgg-style convnets great again. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 13733-13742).

[15] Han, K., Wang, Y., Tian, Q., Guo, J., Xu, C. and Xu, C., 2020. Ghostnet: More features from cheap operations. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 1580-1589).

[16] Wu, Y. and He, K., 2018. Group normalization. In Proceedings of the European conference on computer vision (ECCV) (pp. 3-19).

[17] Carion, N., Massa, F., Synnaeve, G., Usunier, N., Kirillov, A. and Zagoruyko, S., 2020, August. End-to-end object detection with transformers. In European conference on computer vision (pp. 213-229). Cham: Springer International Publishing.

[18] Zhao, Y., Lv, W., Xu, S., Wei, J., Wang, G., Dang, Q., Liu, Y. and Chen, J., 2024. Detrs beat yolos on real-time object detection. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (pp. 16965-16974).

[19] Li, C., Li, L., Jiang, H., Weng, K., Geng, Y., Li, L., Ke, Z., Li, Q., Cheng, M., Nie, W. and Li, Y., 2022. YOLOv6: A single-stage object detection framework for industrial applications. arXiv preprint arXiv:2209.02976.

[20] Wang, C.Y., Bochkovskiy, A. and Liao, H.Y.M., 2023. YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 7464-7475).